

1. TIME RESPONSE ANALYSIS

The transfer function $G(s)$ of any system will be a ratio with numerator and denominator appearing as polynomials in s . Some examples are given in table 1.

Table 1. Examples of transfer function

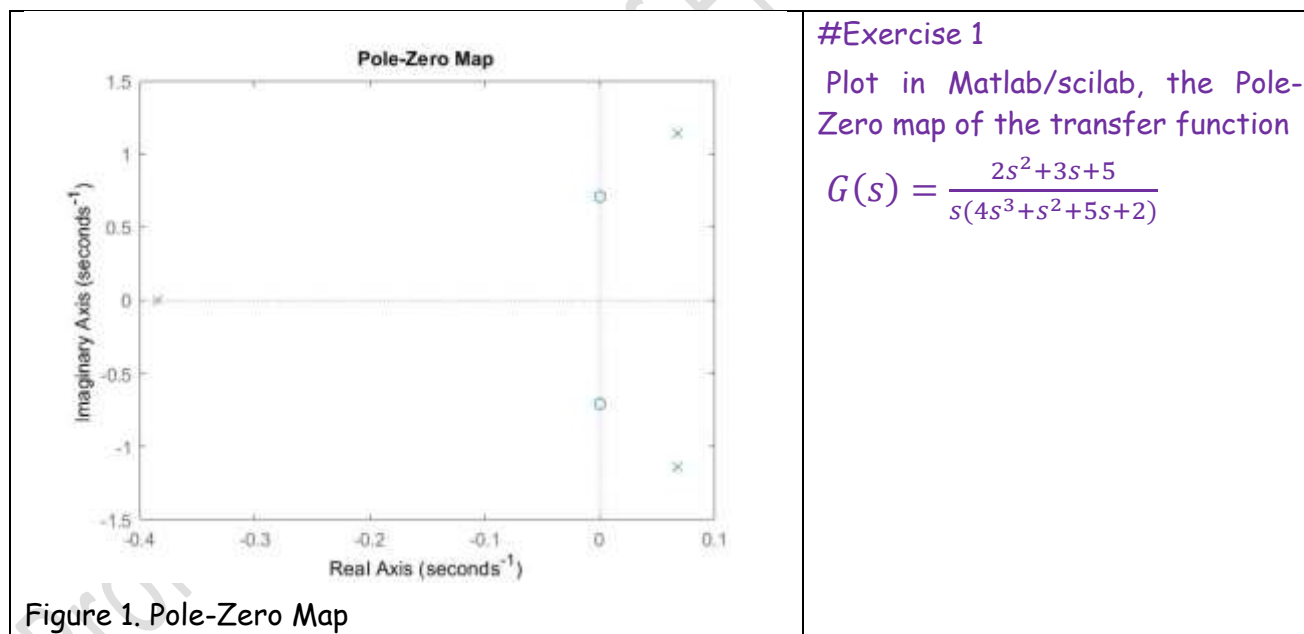
$G(s) = \frac{2}{s+2}$	$G(s) = \frac{2s}{s+2}$	$G(s) = \frac{s+1}{s+5}$	$G(s) = \frac{s+2}{(s+5)(s+1)}$	$G(s) = \frac{2s^2+1}{4s^3+s^2+5s+2}$
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In general, the transfer function can be written as $G(s) = K \frac{p(s)}{q(s)}$ where K is the gain. The degree of $p(s)$ will be always less than the degree of $q(s)$. The equation $q(s)=0$ is known as the **characteristic equation**, and the roots of the characteristic equation $q(s)=0$ are known as the **poles**. The roots of the equation $p(s)=0$ are known as **zeroes** of the system. The poles and zeroes of transfer functions given in Table 1 are shown below in Table 2.

Table 2. Poles and zeroes

Zero	No zero	$s=0$	$s=-1$	$s=-2$	$s=0 \pm j0.707$
Pole	$s=-2$	$s=-2$	$s=-5$	$s=-1, -5$	$s=-0.3842, 0.0671 \pm j1.1389$

The complex zeroes and poles appear as conjugate pairs. The poles and zeroes can be represented in a complex plane (s is a complex variable $\sigma + j\omega$). The poles are represented by 'x' zeroes by 'o'. The pole-zero map of the last transfer function in Table 1 is shown in figure 1.



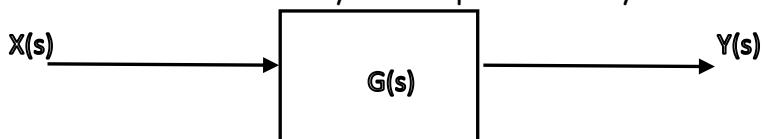
The poles and zeroes influence the transient and steady state responses of the system. We can always represent the transfer function as $G(s) = K \frac{\pi(s+z_i)}{\pi(s+p_i)}$ where K is the gain (constant), where K is the gain (constant), z_i represent the zeroes, p_i represents the poles, and π stands for the product. Hence the transfer function $G(s) = \frac{2s^2+1}{4s^3+s^2+5s+2}$ can be written as

$$G(s) = 0.5 \frac{(s+j0.707)(s-j0.707)}{(s+0.3842)(s-0.0671+j1.1389)(s-0.0671-j1.1389)}$$

The value of $K=0.5$, zeroes $z_i = \pm j0.707$ and poles $p_i = -0.3842, 0.0671 \pm j1.1389$

1.1 I order systems

Consider a first order system represented by the transfer function $G(s) = \frac{Y(s)}{X(s)} = \frac{a}{s+a}$



There is one pole at $s=-a$, and no zeroes for the system. The transfer function can also be written as $G(s) = \frac{1}{1+Ts}$ where $T = \frac{1}{a}$.

1.1.1 Step Response

For a unit step input, $x(t)=1$, $X(s)=1/s$

$Y(s) = \frac{1}{s} \left(\frac{a}{s+a} \right) = \frac{A}{s} + \frac{B}{s+a} = \frac{1}{s} - \frac{1}{s+a}$ (by partial fraction). The response $y(t)$ is obtained by taking the inverse Laplace Transform.

$$y(t) = 1 - e^{-at} = 1 - e^{-\frac{t}{T}}$$

The error $e(t) = \text{Desired output} - \text{actual output} = x(t) - y(t) = 1 - (1 - e^{-at}) = e^{-at}$

When $t = \frac{1}{a}$, $y(t) = 1 - e^{-1} = 0.632$.

The value of the time taken by the system to reach 63.2% of the final output is known as the **time constant** (T) of the system which represents the speed of response of the system. The smaller the time constant of the system the faster the response of the system.

The steady state (final) value of the response is obtained by

$$\lim_{t \rightarrow \infty} y(t) = 1 \quad \text{or} \quad \lim_{s \rightarrow 0} sY(s) = \lim_{s \rightarrow 0} s \frac{1}{s} \left(\frac{a}{s+a} \right) = 1$$

The steady state error is $\lim_{t \rightarrow \infty} e(t) = e^{-at} = 0$

For a particular case of $a=2$ (ie., $G(s) = \frac{2}{s+2} = \frac{1}{1+0.5s}$), the step response $y(t)$ is plotted against time t and is shown in figure 2. The blue line is the input (desired output) and the red is the actual output. The time constant of the system is $T=0.5$ s.

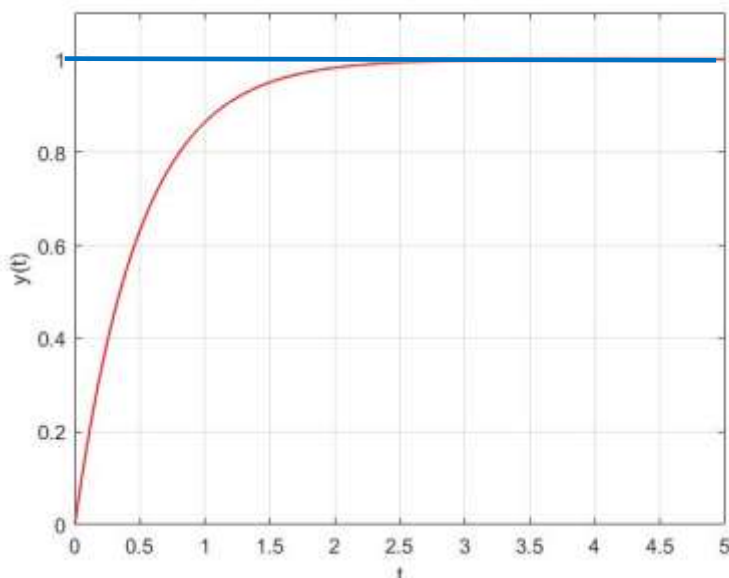


Fig 2. Step response of a first order system

Exercise 2a

Using Matlab/scilab, plot

In one figure the step response of the first order system by taking $a=1, 3$, and 5 and compare the time constants

Exercise 2b

Obtain the expression for the step response of $G(s) = \frac{b}{s+a}$.

Look at the step responses of

$$G(s) = \frac{1}{s+2}, \quad G(s) = \frac{2}{s+2},$$

$G(s) = \frac{3}{s+2}$ by plotting them in the same figure using Matlab/Scilab

1.1.2 Ramp response

For a unit ramp, $x(t)=t$, $X(s)=1/s^2$

$$Y(s) = \frac{1}{s^2} \cdot \frac{a}{s+a} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+a} = \left(\frac{-1}{a}\right) \frac{1}{s} + \frac{1}{s^2} + \left(\frac{1}{a}\right) \frac{1}{s+a} \text{ (by partial fraction)}$$

$$\therefore y(t) = \left(\frac{-1}{a}\right) + t + \left(\frac{1}{a}\right) e^{-at} = t - \frac{1}{a}(1 - e^{-at}) = t - T(1 - e^{-\frac{t}{T}})$$

The unit ramp response of a first order system $G(s) = \frac{2}{s+2}$ is shown in figure 3. Red line is the desired output(input) and blue line is the actual output.

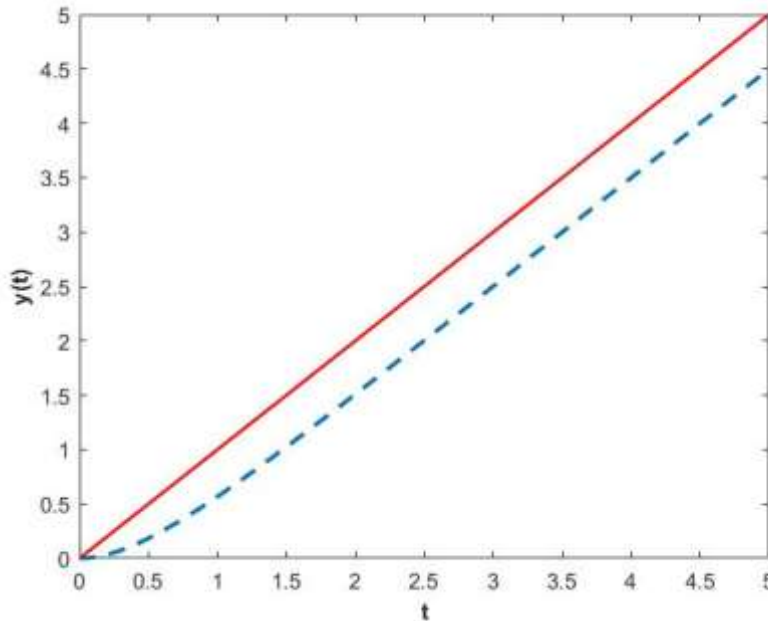


Fig 3. Unit ramp response of a first order system

As t takes a large value, the term $(1 - e^{-at}) \rightarrow 1$. Then $y(t) = t - \frac{1}{a}$. That is, at steady state, the actual output will be always less than the desired output by $(\frac{1}{a})$.

$$\text{The error } e(t) = x(t) - y(t) = t - \left[t - \frac{1}{a}(1 - e^{-at})\right] = \frac{1}{a}(1 - e^{-at})$$

At steady state (as $t \rightarrow \infty$), the error $e(t) = \frac{1}{a}$

1.1.3 Impulse Response

For a unit impulse, $x(t) = \delta(t)$, $X(s)=1$

$$Y(s) = 1 \cdot \frac{a}{s+a}$$

$$y(t) = ae^{-at} = \frac{1}{T} e^{-\frac{t}{T}}$$

As $t \rightarrow \infty$, the response $y(t) = 0$

The unit impulse response of a first order system $G(s) = \frac{2}{s+2}$ is shown in figure 4. The input acts only for a short duration of time and as time progresses, the response exponentially decays to zero.

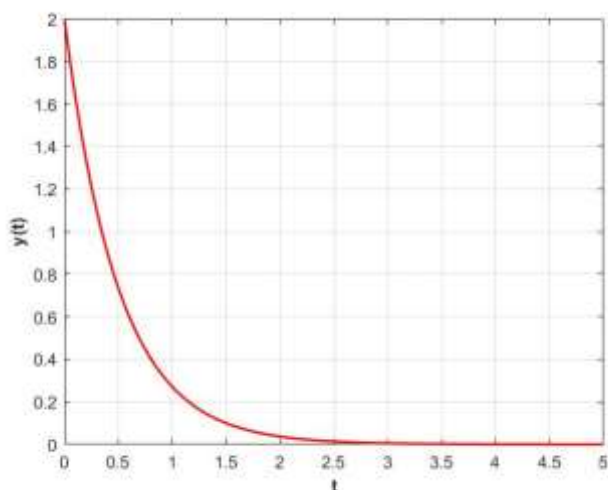


Figure 4. Unit impulse response

#Exercise 3

Using Matlab/Scilab, plot the impulse responses of the following first order systems in the same graph.

1. $G(s) = \frac{1}{s+1}$
2. $G(s) = \frac{5}{s+5}$

Now for a first order system with transfer function $G(s) = \frac{a}{s+a}$,

Ramp response $y(t) = t - \frac{1}{a}(1 - e^{-at})$

Step response $y(t) = 1 - e^{-at} = \frac{d}{dt}(\text{ramp response})$

Impulse response $y(t) = ae^{-at} = \frac{d}{dt}(\text{step response})$

#Exercise 4a

A first order system has a transfer function $G(s) = \frac{125}{100s+125}$. What is the time constant of the system? Determine the output of the system after $t=0.5$ s when it is subjected to a unit step, unit ramp, and unit impulse inputs. What is the steady state error for unit ramp input?

#Exercise 4b

A first order system has a unit step response given by $y(t) = 1 - e^{-0.75t}$. What will be its responses to unit ramp and unit impulse inputs after 1 s. What is the transfer function of the system?

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